

Phillip Macias

Computational Scientist & Data Modeler, Mountain View, CA

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Summary

Computational modeler with 10+ years of experience designing and executing large-scale simulation frameworks in collaborative, deadline-driven environments. Experienced in distributed HPC workflows, high-dimensional parameter exploration, and uncertainty quantification and propagation. Contributed to simulation infrastructure for a NASA flagship mission. Now applying this modeling and statistical background to data science, with recent work in applied machine learning.

Skills

- **Computation:** Python (incl. NumPy, SciPy, Pandas, Matplotlib, Jupyter), SQL, DuckDB, geospatial data (raster/HDF5), Git, reproducible pipelines, HPC (Slurm, distributed simulation campaigns, large-scale parameter sweeps), agentic AI workflows (Claude Code, LLM APIs), Bash, $\text{\LaTeX}$
- **Statistics/Machine Learning:** probability and Monte Carlo methods, uncertainty quantification and propagation, regression and classification (XGBoost, Explainable Boosting Machines), convolutional neural networks (PyTorch, Grad-CAM), feature engineering, data cleaning/wrangling, model explainability (SHAP), data visualization

Experience and Projects

Flashpoint — Wildfire Severity Classifier (Self-Directed Project) 2026

- Built an end-to-end ML pipeline predicting whether an active wildfire escalates or stays contained, using only the first 1–2 days of fire growth, along with weather, topography, and vegetation features.
- Compared multiple modeling approaches (XGBoost, Explainable Boosting Machine, CNN with Grad-CAM) to identify the best-performing method for the classification task, with EBM emerging as the most reliable across held-out years.

Roman Space Telescope Planning (Postdoctoral Researcher, UC Santa Cruz) 2020 – 2026

- Member of Science Investigation Team simulations subgroup: independently ran ~ 500 end-to-end simulation campaigns evaluating candidate survey designs (depth, cadence, filters) against a NASA top-level requirement to improve dark-energy constraining power 10x, applying a standardized metric to translate each design's simulated performance into a single comparable figure-of-merit.
- Synthesized simulation results into a recommended survey design adopted by NASA, which exceeded the requirement (FoM 340 vs. 326 target) and directly shaped the observing strategy for the flagship dark-energy survey of *Roman*, a \$4.3B NASA mission launching August 2026.

Graduate Research (NSF Graduate Research Fellow, UC Santa Cruz) 2012 – 2019

- Numerically investigated accretion flows across a broad range of astrophysical systems, running large hydrodynamic simulations and deriving analytical constraints to reduce the effective parameter space, resulting in 5 co-authorships in computational astrophysics.
- Used high-resolution stellar spectra to place constraints on heavy-element production channels, ruling out 2 leading candidates and resulting in 2 first-author publications in nuclear astrophysics.

Mentorship & Research Output 2012 – 2026

- Mentored **20+** students (high school through graduate level) in computational research workflows.
- Co-authored **21** peer-reviewed publications in theoretical and observational astrophysics.

Education

Ph.D., Astronomy & Astrophysics — University of California, Santa Cruz 2019
M.S., Astronomy & Astrophysics — University of California, Santa Cruz 2015
B.S., Physics (Highest Honors) — University of California, Santa Barbara 2012